CHAPTER 1 PHYSICAL QUANTITIES

Key idea: every quantity is a number × a unit, and every term in a correct equation has the same dimensions.

1 SI BASE QUANTITIES

Quantity	Unit	Dim.
Length	metre, m	[L]
Mass	kilogram, kg	[M]
Time	second, s	[T]
Current	ampere, A	[I]
Temperature	kelvin, K	[θ]
Amount	mole, mol	[N]
Luminous intensity	candela, cd	–

Derived units are products or quotients of base units.

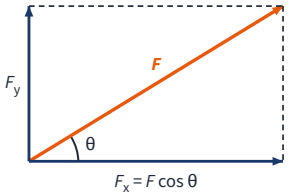
2 DIMENSIONS & HOMOGENEITY

$$[F] = \mathrm{MLT}^{-2}$$

$$[E] = \mathrm{ML}^2\mathrm{T}^{-2}$$

- A correct equation is **homogeneous**: every term has the same dimensions.
- Homogeneous does **not** mean correct: numbers such as ½ or 2π have no dimensions.
- Angles (rad), strain and refractive index are dimensionless.

3 SCALARS & VECTORS



$$F_x = F\cos\theta \qquad F_y = F\sin\theta$$

$$R = \sqrt{R_x^2 + R_y^2} \qquad \tan\theta = \frac{R_y}{R_x}$$

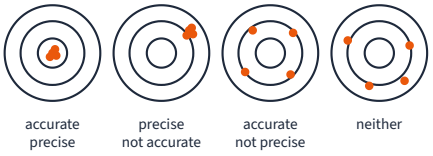
Scalar: magnitude only. **Vector:** magnitude and direction.

4 PREFIXES

Prefix	×	Prefix	×
tera T	10^{12}	centi c	10^{-2}
giga G	10^{9}	milli m	10^{-3}
mega M	10^{6}	micro μ	10^{-6}
kilo k	10^{3}	nano n	10^{-9}
deci d	10^{-1}	pico p	10^{-12}

1 cm^2 = 10^{-4} m^2, 1 cm^3 = 10^{-6} m^3, 1 km h^{-1} = 1/3.6 m s^{-1}

5 ACCURACY & PRECISION



Systematic — same size and sign each time (zero error); affects **accuracy**; correct by calibrating

Random — scatter both ways (reaction time); affects **precision**; reduce by averaging

6 UNCERTAINTY RULES

$Z = A \pm B$	$\Delta Z = \Delta A + \Delta B$
$Z = AB,\ A/B$	$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$
$Z = A^n$	$\frac{\Delta Z}{Z} = \lvert n\rvert \frac{\Delta A}{A}$
$Z = kA$	$\Delta Z = k\,\Delta A$

- Analogue scale: ± half the smallest division
- Digital: ± one in the last digit
- Repeated: $\Delta x = \frac{1}{2}(x_{\max} - x_{\min})$

7 UNIT CHECK: DERIVED QUANTITIES

Quantity	Defining equation	SI base units	Dimensions
Force	$F = ma$	kg m s^{-2} (N)	MLT^{-2}
Energy, work	$W = Fs$	kg m^2 s^{-2} (J)	ML^2T^{-2}
Power	$P = W/t$	kg m^2 s^{-3} (W)	ML^2T^{-3}
Pressure	$p = F/A$	kg m^{-1} s^{-2} (Pa)	$ML^{-1}T^{-2}$
Momentum	$p = mv$	kg m s^{-1}	MLT^{-1}
Density	$\rho = m/V$	kg m^{-3}	ML^{-3}

8 COMMON SECTION A TRAPS

- Taking the newton or joule as a **base** unit; they are derived.
- Adding **percentage** uncertainties for a sum; add absolute ones.
- Forgetting to multiply by the power: $\Delta(r^2)/r^2 = 2\Delta r/r$.
- Quoting the answer to more significant figures than the data.
- Using cm in a formula that needs m.

EXAM TIPS

- ✓ Write the defining equation, then replace each quantity by its base units.
- ✓ For dimensional analysis, equate the powers of M, L and T.
- ✓ Quote the uncertainty to 1 s.f. and the value to the same decimal place.
- ✓ Resolve every vector along two perpendicular axes before adding.

CHAPTER 2 KINEMATICS

Key idea: describe motion with s, v, a and t; for constant acceleration use the four equations of motion.

1 DEFINITIONS

Displacement	straight-line distance in a stated direction (vector)
Velocity	$v = \frac{ds}{dt}$, rate of change of displacement
Acceleration	$a = \frac{dv}{dt}$, rate of change of velocity
Speed, distance	scalars: magnitude only

Average velocity $= \frac{\Delta s}{\Delta t}$; average speed $= \frac{\text{total distance}}{\text{total time}}$

2 EQUATIONS OF MOTION

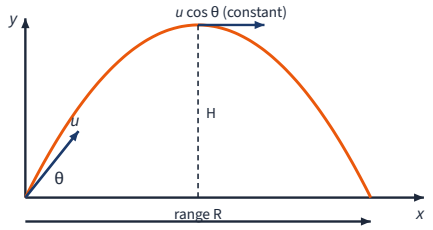
$$v = u + at \qquad s = ut + \tfrac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \tfrac{1}{2}(u + v)t$$

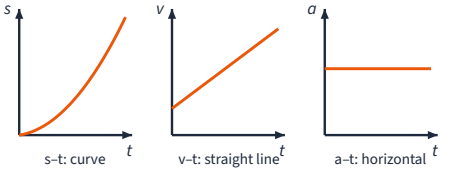
Constant acceleration in a straight line only.

3 PROJECTILE MOTION



Horizontal	constant velocity $u\cos\theta$, $a_x = 0$
Vertical	acceleration g downward, $u_y = u\sin\theta$
Top	$v_y = 0$; v_x unchanged

4 MOTION GRAPHS



Graph	Gradient	Area
s–t	velocity	–
v–t	acceleration	displacement
a–t	–	change in v

Area below the t-axis on a v–t graph is negative displacement.

5 FREE FALL

$$a = -g = -9.81\ \text{m s}^{-2}$$

- Gravity only; air resistance neglected
- Same g for all masses
- Choose **up** as positive and keep the sign
- At the highest point $v = 0$, but $a = g$ still

6 PROJECTILE FORMULAS

$$T = \frac{2u\sin\theta}{g}$$

$$H = \frac{u^2\sin^2\theta}{2g}$$

$$R = \frac{u^2\sin 2\theta}{g}$$

- $R_{\max}$ at $\theta = 45^\circ$; θ and $90^\circ - \theta$ give the same range
- Path is a **parabola**: $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

7 HORIZONTAL LAUNCH FROM HEIGHT H

$$t = \sqrt{\frac{2h}{g}} \qquad x = ut \qquad v = \sqrt{u^2 + (gt)^2}$$

- Time of fall depends on h only, not on u
- Angle below horizontal: $\tan\phi = \frac{gt}{u}$

8 COMMON SECTION A TRAPS

- Mixing signs: if up is +, then g is −9.81 m s^{-2}.
- Using $R = u^2\sin 2\theta/g$ when landing at a **different** height.
- Saying acceleration is zero at the top of the path.
- Treating distance as displacement on a return journey.

EXAM TIPS

- ✓List s, u, v, a, t; pick the equation without the unknown you do not need.
- ✓Split projectile motion into horizontal and vertical parts.
- ✓Time links the two directions.
- ✓State the direction of every vector answer.

CHAPTER 3

DYNAMICS

Key idea: a resultant force changes momentum; in an isolated system the total momentum is conserved.

1 NEWTON'S LAWS

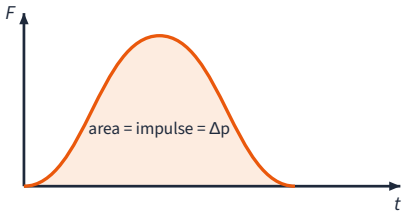
1. **1st:** a body stays at rest or at constant velocity unless a resultant force acts (inertia, measured by mass).
2. **2nd:** rate of change of momentum ∝ resultant force, in its direction.
3. **3rd:** A on B = − B on A; the pair act on **different** bodies.

$$F = \frac{dp}{dt} \qquad F = ma$$

1 N gives 1 kg an acceleration of 1 m s^{-2}.

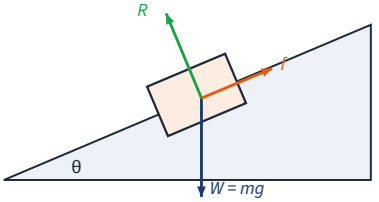
2 MOMENTUM & IMPULSE

$$p = mv \qquad J = F\Delta t = \Delta p$$



- Impulse = **area** under the F–t graph
- Unit: kg m s^{-1} = N s

3 FREE-BODY DIAGRAMS



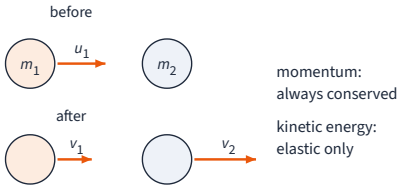
- Smooth incline: $a = g\sin\theta$, $N = mg\cos\theta$
- Lift up: $R = m(g + a)$; down: $R = m(g - a)$
- Free fall: $R = 0$ (weightless)

4 CONSERVATION OF MOMENTUM

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

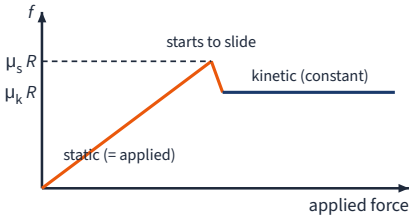
- Isolated system: no resultant external force
- Holds in **every** collision and explosion
- Recoil from rest: $m_1v_1 + m_2v_2 = 0$
- 2-D: conserve p_x and p_y separately

5 COLLISIONS



Type	Momentum	Kinetic energy
Elastic	conserved	conserved
Inelastic	conserved	lost
Perfectly inelastic	conserved	max. loss, stick

6 FRICTION



$$f_s \le \mu_s N \qquad f_k = \mu_k N$$

- Opposes (the tendency of) relative motion
- $\mu_k < \mu_s$; independent of contact area
- About to slide on incline: $\tan\theta = \mu_s$

7 USEFUL RESULTS

Elastic, m_2 at rest:

$$v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2}$$

$$v_2 = \frac{2m_1u_1}{m_1 + m_2}$$

Elastic: speed of approach = speed of separation

$$u_1 - u_2 = v_2 - v_1$$

Kinetic energy: $E_k = \frac{p^2}{2m}$

Pulley (light string): $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$, $T = \frac{2m_1m_2g}{m_1 + m_2}$. Stream on a wall: $F = \frac{dm}{dt}(v - u)$.

8 COMMON SECTION A TRAPS

- Calling weight and normal reaction a 3rd-law pair (same body, different types).
- Rebound: $\Delta p = m(v + u)$, not $m(v - u)$.
- Forgetting momentum is a vector: fix a positive direction.
- Assuming $N = mg$ when the force pulls at an angle.

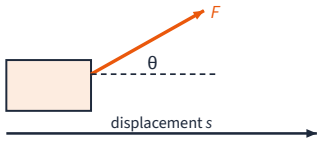
EXAM TIPS

- ✓Draw a free-body diagram for each body first.
- ✓Apply $\Sigma F = ma$ along the direction of motion.
- ✓State "no external force, so momentum is conserved".
- ✓Check kinetic energy to decide whether a collision is elastic.

CHAPTER 4 WORK, ENERGY & POWER

Key idea: work transfers energy; energy is never created or destroyed, only converted.

1 WORK DONE



$$W = Fs\cos\theta$$

- Scalar, unit J (1 J = 1 N × 1 m)
- Variable force: $W = \int F\,dx$ = **area** under F–s
- $\theta = 90^\circ$ (centripetal force, normal reaction): $W = 0$
- Friction does negative work

2 ENERGY STORES

$$E_k = \tfrac{1}{2}mv^2 \qquad \Delta U = mg\Delta h$$

$$U_{\text{spring}} = \tfrac{1}{2}kx^2$$

Elastic energy = area under the F–x graph.

3 POWER

$$P = \frac{W}{t} = \frac{dW}{dt} \qquad P = Fv$$

- Rate of doing work, unit W = J s^{-1}
- Constant speed against resistance F: $P = Fv$
- 1 kW h = 3.6 × 10^6 J

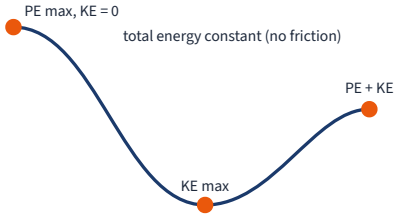
4 WORK–ENERGY THEOREM

$$W_{\text{net}} = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2$$

Net work done on a body = change in its kinetic energy.

Against friction on a level surface: $W = \mu mgs$.

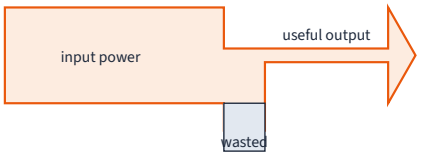
5 CONSERVATION OF ENERGY



$$\tfrac{1}{2}mu^2 + mgh_1 = \tfrac{1}{2}mv^2 + mgh_2 + F_f d$$

- No friction: mechanical energy constant
- From rest through height h: $v = \sqrt{2gh}$, any smooth path
- Pendulum: $v = \sqrt{2gL(1-\cos\theta)}$

6 EFFICIENCY



$$\eta = \frac{\text{useful output}}{\text{input}} \times 100\%$$

The rest is wasted, mostly as heat; $\eta < 100\%$.

7 CONSERVATIVE OR NOT?

Conservative	work independent of path (gravity, spring); energy recoverable
Non-conservative	work depends on path (friction, air resistance); energy → heat

Lifting through height h: $W = mgh$ by any route.

8 COMMON SECTION A TRAPS

- Using the full force instead of its component along s.
- Taking $P = Fv$ with the **driving** force when speed changes; use the net force only for ΔE_k.
- Forgetting the friction term in an energy equation.
- Mixing kW h and J.

EXAM TIPS

- ✓Write "loss of PE = gain in KE + work against friction".
- ✓Use energy methods when the path is curved.
- ✓Use $P = Fv$ for vehicles at constant speed.
- ✓Give efficiency as a percentage or a fraction, consistently.

CHAPTER 5 CIRCULAR MOTION

Key idea: a resultant force towards the centre keeps changing the direction of the velocity.

1 ANGULAR QUANTITIES

$$\theta = \frac{s}{r} \qquad \omega = \frac{d\theta}{dt}$$

$$\omega = \frac{2\pi}{T} = 2\pi f \qquad v = r\omega$$

1 radian: the angle subtended at the centre by an arc equal in length to the radius.

$360° = 2\pi$ rad; 1 rev $s^{-1} = 2\pi$ rad s^{-1}

2 CENTRIPETAL ACCELERATION

$$a = \frac{v^2}{r} = r\omega^2$$

- Directed towards the **centre**
- Speed constant, velocity changes direction, so the body accelerates

3 CENTRIPETAL FORCE

$$F = \frac{mv^2}{r} = mr\omega^2$$

- Not a new force: provided by tension, friction, gravity or normal reaction
- Perpendicular to v: does **no work**; E_k constant
- Removed: body moves off along the **tangent**

4 HORIZONTAL CIRCLES

Level bend (friction provides the force):

$$\mu mg \geq \frac{mv^2}{r} \qquad v_{\max} = \sqrt{\mu g r}$$

Banked track, no friction:

$$\tan\theta = \frac{v^2}{rg}$$

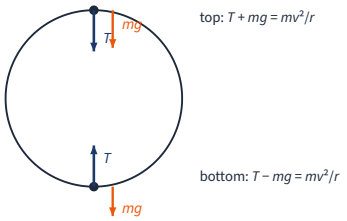
Conical pendulum:

$$T\cos\theta = mg$$

$$T\sin\theta = \frac{mv^2}{r}$$

Resolve vertically (balance) and horizontally (centripetal).

5 VERTICAL CIRCLE

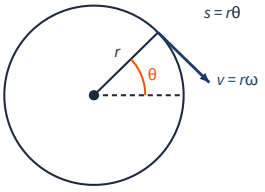


$$\text{Top: } T + mg = \frac{mv^2}{r}$$

$$\text{Bottom: } T - mg = \frac{mv^2}{r}$$

- Tension is greatest at the **bottom**
- Just completes the loop: $T = 0$ at top, $v_{\text{top}} = \sqrt{gr}$
- Convex bridge top: $mg - N = \frac{mv^2}{r}$

6 UNIFORM CIRCULAR MOTION



Quantity	Constant?
speed, ω, E_k	yes
velocity	no (direction)
acceleration	no (direction)

7 UNIT CHECK

Quantity	Formula	Unit
Angular displacement	$\theta = s/r$	rad
Angular velocity	$\omega = 2\pi f$	rad s^{-1}
Centripetal acceleration	v^2/r	m s^{-2}
Centripetal force	$mr\omega^2$	N

8 COMMON SECTION A TRAPS

- Adding a "centrifugal force" to the free-body diagram.
- Using degrees instead of radians in $s = r\theta$.
- Saying velocity is constant (only speed is).
- Using the diameter instead of the radius.
- At the top, forgetting that T and mg both point down.

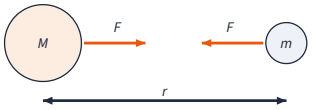
EXAM TIPS

- ✓Draw the forces, then mark the centre of the circle.
- ✓Net force towards centre $= mv^2/r$.
- ✓Use energy conservation to link top and bottom speeds.
- ✓Convert rev min^{-1} to rad s^{-1} first.

CHAPTER 6 GRAVITATION

Key idea: every mass attracts every other mass; the force falls off as $1/r^2$.

1 NEWTON'S LAW OF GRAVITATION



$$F = \frac{GMm}{r^2}$$

- Always attractive; r between centres
- $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
- Equal and opposite on the two masses

2 FIELD STRENGTH G

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

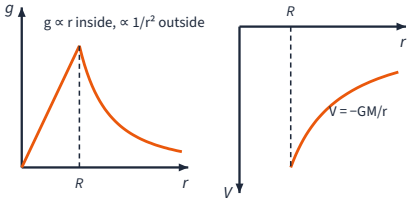
- Force per unit mass, N kg^{-1} (= m s^{-2})
- Surface: $g_0 = \frac{GM}{R^2}$, so $GM = g_0 R^2$
- Height h: $g = \frac{GM}{(R+h)^2}$

3 GRAVITATIONAL POTENTIAL

$$V = -\frac{GM}{r} \qquad U = -\frac{GMm}{r}$$

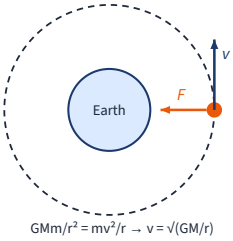
- Work done per unit mass to bring a mass from **infinity**; unit J kg^{-1}
- Zero at infinity, negative elsewhere
- $g = -\frac{dV}{dr}$: field = −(potential gradient)
- Moving m: $W = m\Delta V$

4 INSIDE AND OUTSIDE



- Inside (uniform Earth): $g \propto r$
- Outside: $g \propto 1/r^2$; $V \propto -1/r$
- $g = 0$ at the centre

5 SATELLITE MOTION



$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}} \qquad T^2 = \frac{4\pi^2}{GM} r^3$$

Lower orbit: faster, shorter period.

6 ENERGY OF A SATELLITE

	Expression
Kinetic	$+\frac{GMm}{2r}$
Potential	$-\frac{GMm}{r}$
Total	$-\frac{GMm}{2r}$

$$v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2g_0 R}$$

Escape speed does not depend on the mass of the object.

7 GEOSTATIONARY ORBIT

- Period 24 h, in the **equatorial plane**
- Moves west to east with the Earth: stays above one point
- Radius about 4.2×10^7 m (height ≈ 36 000 km)
- Astronauts feel weightless: they are in **free fall**, not free of gravity

8 COMMON SECTION A TRAPS

- Using the height h instead of $r = R + h$.
- Dropping the minus sign in V and U.
- Saying $g = 0$ in orbit.
- Thinking escape speed depends on the projectile mass.
- r doubled: F becomes $\frac{1}{4}$, V becomes $\frac{1}{2}$.

EXAM TIPS

- ✓Start satellite questions with "gravitational force provides the centripetal force".
- ✓Replace GM by $g_0 R^2$ when G and M are not given.
- ✓Use ratios for "how many times" questions.
- ✓Energy needed between orbits = ΔE_{total}.

CHAPTER 7

ROTATION OF RIGID BODIES

Key idea: torque, moment of inertia and angular momentum play the roles of force, mass and momentum.

1 ROTATIONAL KINEMATICS

$$\omega = \omega_0 + \alpha t$$

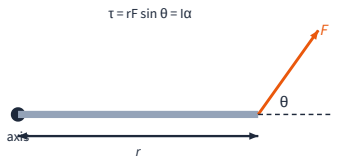
$$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

Link to the rim:

$s = r\theta$ $v = r\omega$ $a = r\alpha$

2 TORQUE



$\tau = rF\sin\theta$ $\tau = I\alpha$

Unit N m; only the perpendicular component turns the body.

3 MOMENT OF INERTIA

$$I = \sum mr^2$$

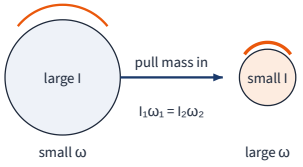
Body (axis)	I
Ring, hoop	MR^2
Disc, cylinder	$\frac{1}{2}MR^2$
Solid sphere	$\frac{2}{5}MR^2$
Rod, centre	$\frac{1}{12}ML^2$
Rod, end	$\frac{1}{3}ML^2$

Depends on mass, its distribution and the axis.

4 LINEAR ↔ ROTATIONAL

Linear	Rotational
s	θ
v, a	ω, α
m	I
$F = ma$	$\tau = I\alpha$
$p = mv$	$L = I\omega$
$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$
$W = Fs, P = Fv$	$W = \tau\theta, P = \tau\omega$

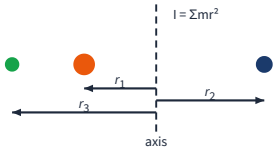
5 ANGULAR MOMENTUM



$L = I\omega$ $\tau = \dfrac{dL}{dt}$

Conservation: no external torque, $I_1\omega_1 = I_2\omega_2$ (skater pulls arms in: I ↓, ω ↑).

6 ROTATIONAL KE & ROLLING



$$E_k = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2$$

Rolling without slipping: $v = r\omega$.

7 ROLLING DOWN A SLOPE FROM REST

$$mgh = \tfrac{1}{2}mv^2\left(1 + \frac{I}{mr^2}\right)$$

Body	v^2
Ring	gh
Disc	$\frac{4}{3}gh$
Sphere	$\frac{10}{7}gh$

Smaller I/mr^2 reaches the bottom first: sphere, then disc, then ring.

8 COMMON SECTION A TRAPS

- Using rev or degrees: convert to rad (1 rev = 2π rad).
- Forgetting the rotational KE of a rolling body.
- Using I about the centre when the axis is at the end.
- Claiming kinetic energy is conserved when the skater pulls in (it increases).

EXAM TIPS

✓Write $\tau = I\alpha$ for each rotating body and $F = ma$ for each hanging mass.

✓Link them with $a = r\alpha$.

✓Use angular momentum when the shape changes.

✓Use energy when heights are given.

CHAPTER 8 STATICS

Key idea: a body is in equilibrium when the resultant force and the resultant torque are both zero.

1 CONDITIONS OF EQUILIBRIUM

$$\sum F_x = 0 \qquad \sum F_y = 0$$

$$\sum \tau = 0 \text{ about any point}$$

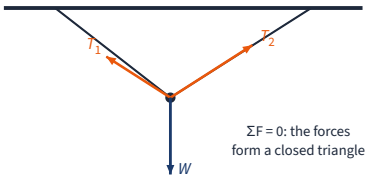
- First: translational (no linear acceleration)
- Second: rotational (no angular acceleration)

2 MOMENT OF A FORCE

$$\text{moment} = F \times d_\perp$$

- $d_\perp$: perpendicular distance from the pivot to the **line of action**
- Unit N m
- **Couple:** two equal, opposite, parallel forces; torque $= F\times$ separation; no resultant force

3 THREE FORCES



- Lines of action **concurrent** (meet at a point)
- Form a closed triangle, head to tail

$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Lami's theorem: angle opposite each force.

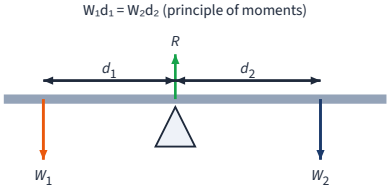
4 CENTRE OF GRAVITY

The point through which the whole **weight** appears to act.

$$\bar{x} = \frac{\sum mx}{\sum m}$$

- Uniform regular body: at its geometric centre
- Suspended freely: lies vertically below the pivot
- Same as centre of mass in a uniform field

5 PRINCIPLE OF MOMENTS



Σ clockwise moments = Σ anticlockwise moments about the same point.

Take moments about the point with the **most unknown** forces.

6 STABILITY & TOPPLING

Stable	CG rises when tilted; returns
Unstable	CG falls; moves away
Neutral	CG height unchanged

- Topples when the vertical through the CG falls **outside the base**
- Low CG and wide base give stability
- Block on a slope topples when $\tan\theta > w/h$

7 METHOD: LADDER, BEAM, HINGE

1. Draw every force: weight at the CG, reactions, friction, tension.
2. Resolve horizontally and vertically: $\sum F_x = 0, \sum F_y = 0$.
3. Take moments about a point that removes an unknown.
4. Hinge force: find both components, then $R = \sqrt{R_x^2 + R_y^2}$.

8 COMMON SECTION A TRAPS

- Using the length of the rod instead of the **perpendicular** distance.
- Forgetting the weight of the beam itself.
- Smooth wall: reaction is **perpendicular** to the wall, no friction.
- Thinking a couple can be balanced by a single force.

EXAM TIPS

- ✓Label the pivot and every distance on the diagram.
- ✓State the principle of moments in words before using it.
- ✓Check: upward forces = downward forces.
- ✓Give the direction of a hinge force as an angle.

CHAPTER 9

DEFORMATION OF SOLIDS

Key idea: within the limit of proportionality, stress is proportional to strain.

1 STRESS & STRAIN

$$\sigma = \frac{F}{A} \qquad \varepsilon = \frac{x}{L_0}$$

Stress normal force per unit area, Pa (N m^{-2})

Strain extension ÷ original length, **no unit**

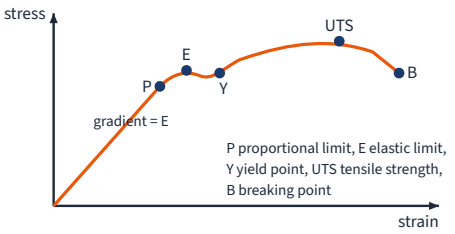
Wire of diameter d: $A = \dfrac{\pi d^2}{4}$

2 YOUNG'S MODULUS

$$E = \frac{\sigma}{\varepsilon} = \frac{FL_0}{Ax}$$

- Gradient of the linear part of the σ–ε graph
- Unit Pa; a property of the **material**, not of the sample

3 STRESS–STRAIN GRAPH



P limit of proportionality
E elastic limit
Y yield point: plastic flow
UTS maximum stress
B breaking point

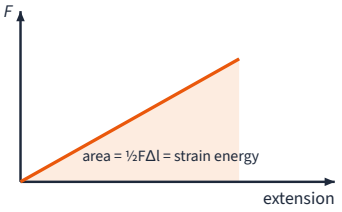
4 HOOKE'S LAW

$$F = kx \qquad k = \frac{EA}{L_0}$$

Extension ∝ force, provided the limit of proportionality is not exceeded.

- Elastic: returns to original length
- Plastic: permanent extension remains

5 ELASTIC POTENTIAL ENERGY



$$U = \tfrac{1}{2}Fx = \tfrac{1}{2}kx^2$$

$$\frac{U}{V} = \tfrac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2E}$$

Energy = **area** under the F–x graph.

6 TYPES OF MATERIAL

Ductile large plastic region before breaking (copper, mild steel)

Brittle breaks just after the elastic region (glass, cast iron)

Rubber large strain; unloading curve below loading: area between = heat lost (hysteresis)

7 UNIT CHECK

Quantity	Formula	Unit
Stress	F/A	Pa
Strain	x/L_0	–
Young modulus	σ/ε	Pa
Force constant	EA/L_0	N m^{-1}
Energy density	$\frac{1}{2}\sigma\varepsilon$	J m^{-3}

8 COMMON SECTION A TRAPS

- Using the diameter as the radius in $A = \pi r^2$.
- Forgetting mm → m for extension and diameter.
- Using $U = Fx$ instead of $\frac{1}{2}Fx$.
- Thinking a thicker wire of the same metal has a larger E (it has a larger k).
- Applying $\frac{1}{2}Fx$ beyond the limit of proportionality.

EXAM TIPS

- ✓Find E from the gradient of the straight part of the graph.
- ✓Give strain as a pure number, e.g. 2.0×10^{-3}.
- ✓Use ratios when two wires of the same material are compared.
- ✓Area under a curve: count squares or use trapezia.

CHAPTER 10 GASES

Key idea: gas pressure comes from molecules hitting the walls; temperature measures their mean kinetic energy.

1 IDEAL GAS EQUATION

$$pV = nRT = NkT$$

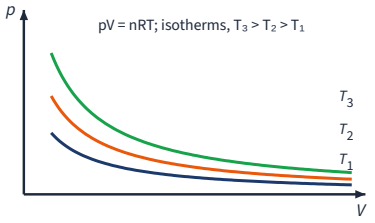
- R 8.31 J K^{-1} mol^{-1}
- k $R/N_A = 1.38 \times 10^{-23}$ J K^{-1}
- N_A 6.02×10^{23} mol^{-1}
- n $= \dfrac{m}{M} = \dfrac{N}{N_A}$

T in kelvin: $T = \theta + 273$

2 GAS LAWS

Law	Constant	Relation
Boyle's	T	pV = constant
Charles's	p	$V \propto T$
Pressure	V	$p \propto T$

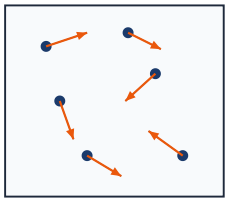
$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$$



3 KINETIC THEORY ASSUMPTIONS

1. Very many identical molecules in **random** motion
2. Volume of molecules negligible
3. No forces except during collisions
4. Collisions are **elastic**; collision time negligible
5. Newtonian mechanics applies

4 PRESSURE OF A GAS



$$p = \tfrac{1}{3}\rho\langle c^2\rangle = \tfrac{1}{3}\frac{Nm}{V}\langle c^2\rangle$$

Pressure = rate of change of momentum of molecules per unit area of wall.

5 MOLECULAR SPEEDS

$$c_{\text{rms}} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

- $c_{\text{rms}} \propto \sqrt{T}$ and $\propto 1/\sqrt{M}$
- Mean square: square first, then average

6 ENERGY & EQUIPARTITION

$$\langle E_k\rangle = \tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

Each degree of freedom: $\frac{1}{2}kT$ per molecule.

Gas	f	U for n mol
Monatomic	3	$\frac{3}{2}nRT$
Diatomic	5	$\frac{5}{2}nRT$
Polyatomic	6	$3nRT$

7 UNIT CHECK

Quantity	Unit
Pressure p	Pa = N m^{-2}
Molar mass M	kg mol^{-1} (not g mol^{-1})
$\langle c^2\rangle$	m^2 s^{-2}
Internal energy U	J

8 COMMON SECTION A TRAPS

- Using °C instead of K.
- Using M in g mol^{-1} in $\sqrt{3RT/M}$.
- Confusing n (moles) with N (molecules).
- Mean KE depends only on T, not on the type of gas.
- Doubling T multiplies c_{rms} by $\sqrt{2}$, not 2.

EXAM TIPS

- ✓Convert cm^3 → m^3 (×10^{-6}) and kPa → Pa.
- ✓Write $n = m/M$ before using $pV = nRT$.
- ✓Use ratios: $c_{\text{rms}} \propto \sqrt{T/M}$.
- ✓List the assumptions in full sentences.

CHAPTER 11 THERMODYNAMICS

Key idea: heat supplied = increase in internal energy + work done by the gas.

1 FIRST LAW

$$\Delta U = Q - W$$

Symbol	+ when	− when
Q	heat into gas	heat out
W	gas expands	gas compressed
ΔU	T rises	T falls

2 INTERNAL ENERGY

$$\Delta U = nC_V\Delta T = \tfrac{f}{2}nR\Delta T$$

- Ideal gas: random KE only, so U depends on T only
- True for **any** process between the same temperatures
- Closed cycle: $\Delta U = 0$

3 WORK DONE BY A GAS

$$W = \int p\,dV \qquad W = p\Delta V$$

- W = **area** under the p–V graph
- Constant pressure: $W = p\Delta V$
- Cycle: net work = area **enclosed** (clockwise: done by gas)

4 MOLAR HEAT CAPACITIES

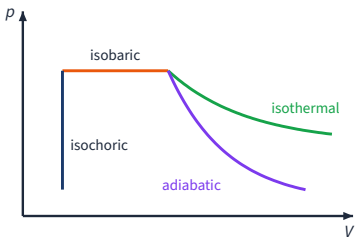
$$C_V = \tfrac{f}{2}R \qquad C_p = C_V + R$$

$$\gamma = \frac{C_p}{C_V}$$

Gas	C_V	C_p	γ
Monatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	5/3
Diatomic	$\frac{5}{2}R$	$\frac{7}{2}R$	7/5

$C_p > C_V$: extra heat does work as the gas expands.

5 THE FOUR PROCESSES



Process	Condition	First law
Isothermal	$\Delta T = 0$	$Q = W$
Adiabatic	$Q = 0$	$\Delta U = -W$
Isobaric	$\Delta p = 0$	$Q = \Delta U + p\Delta V$
Isochoric	$\Delta V = 0$	$Q = \Delta U$

6 ISOTHERMAL VS ADIABATIC

ISOTHERMAL

$$pV = \text{constant}$$

$$W = nRT\ln\frac{V_2}{V_1}$$

ADIABATIC

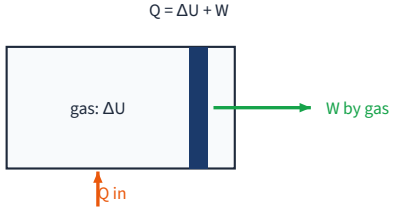
$$pV^\gamma = \text{constant}$$

$$TV^{\gamma-1} = \text{constant}$$

$$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$$

Adiabatic curve is **steeper** ($\gamma > 1$); adiabatic expansion cools the gas.

7 ENERGY FLOW PICTURE



Isothermal: slow, good thermal contact. Adiabatic: fast, insulated.

8 COMMON SECTION A TRAPS

- Writing $\Delta U = Q + W$ with W done **by** the gas.
- Using $nC_p\Delta T$ for ΔU; it is always $nC_V\Delta T$.
- Taking $Q = 0$ as $\Delta T = 0$ (that is isothermal).
- Forgetting kPa → Pa or L → m³ in $p\Delta V$.
- Reading the area enclosed with the wrong sign for an anticlockwise cycle.

EXAM TIPS

- ✓Make a table of Q, W, ΔU for each stage of a cycle.
- ✓Check: around a cycle $\sum \Delta U = 0$.
- ✓Find the missing temperature with $pV = nRT$.
- ✓State the sign convention you use.

CHAPTER 12

TRANSFER OF THERMAL ENERGY

Key idea: heat flows from hot to cold by conduction, convection and radiation.

1 THREE MODES

- **Conduction:** through a material without bulk movement, by lattice vibrations and, in metals, free electrons
- **Convection:** bulk movement of a fluid caused by density differences
- **Radiation:** electromagnetic waves (mainly infrared); no medium needed

2 CONDUCTION

$$\frac{Q}{t} = -kA\frac{d\theta}{dx} = \frac{kA(\theta_1 - \theta_2)}{L}$$

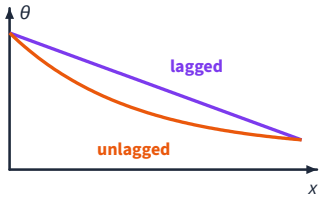
- k: thermal conductivity, W m^{-1} K^{-1}
- $d\theta/dx$: temperature gradient, K m^{-1}
- Minus sign: heat flows down the gradient

3 THERMAL RESISTANCE

$$R = \frac{L}{kA} \qquad \frac{Q}{t} = \frac{\Delta\theta}{R}$$

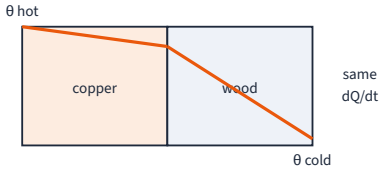
- Layers in series: $R = R_1 + R_2 + \ldots$
- Steady state: **same** rate of heat flow through every layer
- Unit K W^{-1}

4 LAGGED & UNLAGGED BARS



- Lagged: no heat lost from the sides, uniform gradient (straight line)
- Unlagged: heat lost from the sides, gradient decreases along the bar

5 COMPOSITE WALL



$$\frac{k_1 A(\theta_1 - \theta)}{L_1} = \frac{k_2 A(\theta - \theta_2)}{L_2}$$

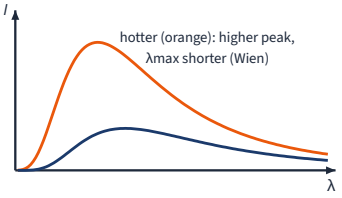
Solve for the interface temperature θ first.

6 RADIATION

$$P = e\sigma AT^4$$

$$P_{\text{net}} = e\sigma A(T^4 - T_0^4)$$

- $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}
- Black body: perfect absorber and emitter, $e = 1$
- T **must** be in kelvin



7 STARS AND PLANETS

$$A = 4\pi r^2 \qquad I = \frac{P}{4\pi d^2}$$

- Intensity at distance d from a source of power P
- Equilibrium: power absorbed = power emitted

8 COMMON SECTION A TRAPS

- Using °C in T^4; a temperature **difference** may be in °C or K.
- Using total length instead of each layer thickness.
- Different rates in series layers: in steady state they are equal.
- Saying radiation needs a medium.

EXAM TIPS

- ✓Draw the temperature profile through a composite wall.
- ✓The layer with the smaller k has the steeper gradient.
- ✓Use ratios for $P \propto T^4$: doubling T gives 16 times the power.
- ✓Quote k with unit W m^{-1} K^{-1}.